

Melina Delgado

SOFTWARE ENGINEER

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Skills

Programming	Go/Golang, Python, Ruby, Bash, C, Scala
DevOps/Cloud	Docker, Kubernetes, AWS, Unix/Linux, Istio, Terraform, AWS CDK, Azure
Tools and Observability	Git, GitHub Actions, Prometheus, Grafana, tcpdump/Wireshark, Splunk, Yara
Databases	AWS DynamoDB, Redis, SQL, MongoDB

Work Experience

Cisco Systems, Talos Intelligence Group

RTP, NC

SOFTWARE ENGINEER

Jan 2022 - Present

Lead software engineer for the *Talos Antispam Logic Engine* (TALN), an AWS **Kubernetes** email scanning service which detects spam, ham, and phishing emails, providing intelligence for the Cisco Secure Email Threat Defense product, which scans **30 million** emails daily and receives up to **1100 messages/second**.

- Re-architected TALN from Scala to Go and into separate Kubernetes micro-services, **increasing application efficiency by 48%**
- Led the TALN service's migration from Azure to AWS, including a storage re-design from file-based mounted volumes to **AWS S3**, a move to **AWS Secrets Manager, AWS SNS/SQS, IAM, Python-based Lambdas**
- Leveraged Istio for security features such as TLS termination, load balancing, and traffic mirroring
- Built *TALN Updater* in Go, which propagates new e-mail scanning rules to hundreds of distributed TALN Kubernetes pods **every 5-15 minutes** to ensure best possible efficacy

Cisco Systems, Security & Trust Organization

RTP, NC

SOFTWARE ENGINEER

May 2018 - Jan 2022

Back-end engineer for *Corona*, which scans Cisco products to generate Software Bill of Materials (SBOM) used in vulnerability and licensing detection, supporting a monthly average of **1.3k users, 12k scans, and 1.5 TB of files scanned**.

- Designed REST APIs for new user-driven features, and implemented bug fixes in Ruby on Rails
- Broke out the scanning capabilities of the monolithic app into Python micro-services using **AWS Batch and EC2**
- Designed file scanners, checksum-based and signature-based (with Yara), for detecting internal Cisco security modules

Yale University, Student Developer & Mentorship Program

New Haven, CT

PROJECT MANAGER AND WEB DEVELOPER

Dec 2016 - May 2019

Developer for the *Vesta* application servicing the Yale housing lottery and room draw

- Re-designed the training curriculum for new Ruby on Rails developers, wrote Rspec unit tests, and provided code review

Education

Yale University

New Haven, CT

BACHELOR'S IN COMPUTER SCIENCE, 3.4 GPA

Aug 2015 - May 2019

Relevant coursework: CPSC 323 Introduction to Systems Programming

Certifications & Courses

May 2020 **GIAC Security Essentials Certification (GSEC)**, SANS Institute

May 2019 **SEC 401: Security Essentials Bootcamp**, SANS Institute

Volunteering

Volunteer Income Tax Assistance and Tax Consulting for the Elderly (VITA/TCE)

New Haven, CT

CO-PRESIDENT, CERTIFIED ADVANCED TAX PREPARER AND CONSULTANT

2016 - 2019

Consult low-income, elderly, ESL, and disabled taxpayers for income tax return preparation

- Serviced over 1,000 taxpayers and co-managed a team of 55+ volunteers
- Cumulatively earned taxpayers around \$1.6 million in refunds (2017)
- Cumulatively saved taxpayers around \$200,000 in tax preparation fees (2017)